Nael Fessha

470-886-2009 | nfessha@gmail.com | https://www.linkedin.com/in/nael-fessha-07003827b/ | naelfessha.github.io

EDUCATION

Georgia Institute of Technology Atlanta, GA
Bachelor of Science in Electrical Engineering GPA: 3.30 Expected May 2028
Relevant Coursework: Programming for HW/SW, Digital Design, Differential Equations

Georgia State University Atlanta, GA
Bachelor of Science in Computer Science GPA: 3.74 Aug 2024 – May 2025
Awards/Honors: Presidents List
Relevant Coursework: Introduction to Python

TECHNICAL SKILLS

Programming Languages: Java, JavaScript, Python, HTML, C, RISC-V Assembly
Software: Excel, PowerPoint, Word, and Power BI, GitHub, MS, Arduino IDE, KiCad 9.0
Hardware/Tools: FPGA design, logic circuits, multimeters, Oscilloscopes, Voltmeters / multimeters, Transformers / variable transformers, Power meters

WORK EXPERIENCE

Georgia Institute of Technology **Atlanta, GA**
Student Assistant Jan 2025 – Present

- Coordinated and organized student seating arrangements to ensure an efficient and distraction-free testing environment.
- Scheduled and managed multiple examinations across different time slots to maintain smooth testing operations.

Front-End Worker | Chick-fil-A Duluth, GA
Shift Leader *Apr 2023- Aug 2023*

- Resolved customer concerns and order discrepancies with professionalism, contributing to the team's goal of maintaining a 93% customer satisfaction rating.
- Trained and mentored 5 new team members on front-end procedures and customer service standards, resulting in their successful integration into the team

PROJECTS

Temperature Controlled Fan (C/C++, Arduino IDE)

- Designed and programmed an Arduino-based temperature monitoring system using a DHT11 sensor and PWM-controlled fan.
- Integrated LCD display output, sensor data acquisition, and motor driver control in a closed-loop embedded system.
- Implemented **C++ firmware** to automatically adjust fan speed based on temperature thresholds

Pathfinding Optimization Engine (C, RISC-V Assembly)

- Engineered an iterative, non-recursive solution using 5-bit binary masking as a hardware-level multiplexer to evaluate 128 path permutations without branch overhead
- Conducted user experience research to assess pressing unmet needs that would drive critical features
- Optimized spatial and temporal performance at the ISA level, minimizing dynamic execution length, static code size, and operand storage requirements against strict baseline metrics.

LEADERSHIP AND PROFESSIONAL DEVELOPMENT

Management Leadership for Tomorrow Atlanta, GA
Career Preparation Fellow Sep 2025 – Present

- Accepted into a selective 20-month professional development program for high-achieving diverse talent

Caleb Foundation-Save Life with Pennies Atlanta, GA
Lead Volunteer Aug 2019 – Present

- Contributed to a program serving 60 children by assisting with meal preparation to support daily operations.
- Secured $22,640 through volunteering at UGA Sports Events in donations as part of a fundraising initiative that enrolled 700 children in school.

ADDITIONAL SKILLS AND INTERESTS

Interested in Embedded Systems, Circuit Debugging, entrepreneurship, and international travel